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PROJECT REPORT

Submitted by-

K M Istiaque - 200021220

MD Wahiduzzaman - 200021228

M M Sadman Shafi - 200021240

Shafayet Sadik Sowad - 200021248

Lossless Image Compression App

Introduction:

In the digital universe where photos hold our cherished memories, we set out on a mission to create a tool that preserves their beauty while reducing their size. Using MATLAB's powerful tools, we crafted an algorithm that delicately balances compression and quality, incorporating masking matrices and discrete cosine transforms. Our goal was to offer users a simple yet effective way to compress images without losing their essence. This project report showcases our journey, highlighting our contribution to the realm of lossless image compression with an emphasis on accessibility and functionality.

Overview of the project:

The project focused on the implementation of lossless image compression techniques using MATLAB, with an emphasis on the integration of masking matrices to achieve optimal compression ratios. Our primary aim was to develop an efficient lossless image compression algorithm in MATLAB, integrating masking techniques to preserve image quality while reducing file sizes.

Throughout the project, we meticulously implemented lossless compression algorithms using MATLAB, leveraging its extensive array of functions and tools. Our focus was particularly on the integration of masking techniques, which play a crucial role in selectively retaining essential image information while discarding redundant data. By exploring various masking matrices and customizing compression parameters, we aimed to achieve maximum compression efficiency without compromising image quality.

Methodology:

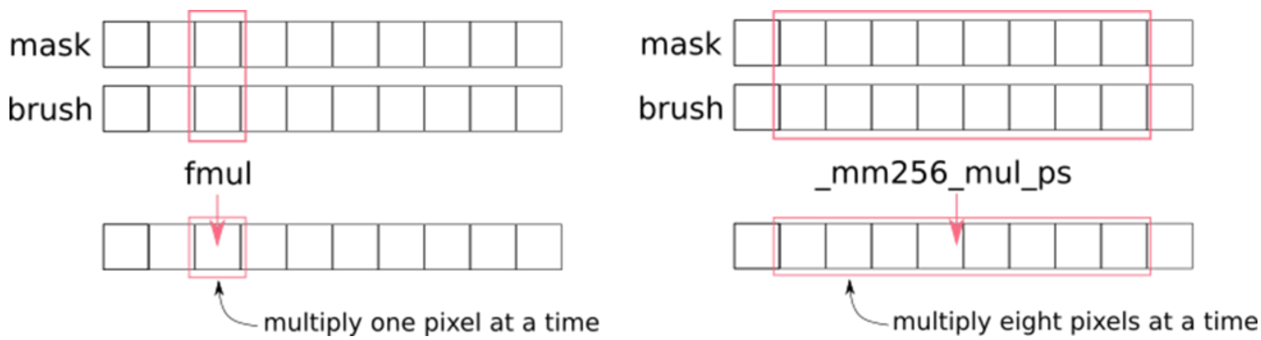
The methodology employed in our project involved several key steps to develop and validate the lossless image compression algorithm.

Implementation of Lossless Compression Algorithms:

We began by implementing lossless compression algorithms using MATLAB, focusing on the integration of masking techniques. By utilizing MATLAB's powerful functions and tools, we developed algorithms capable of compressing image files while preserving their visual fidelity.

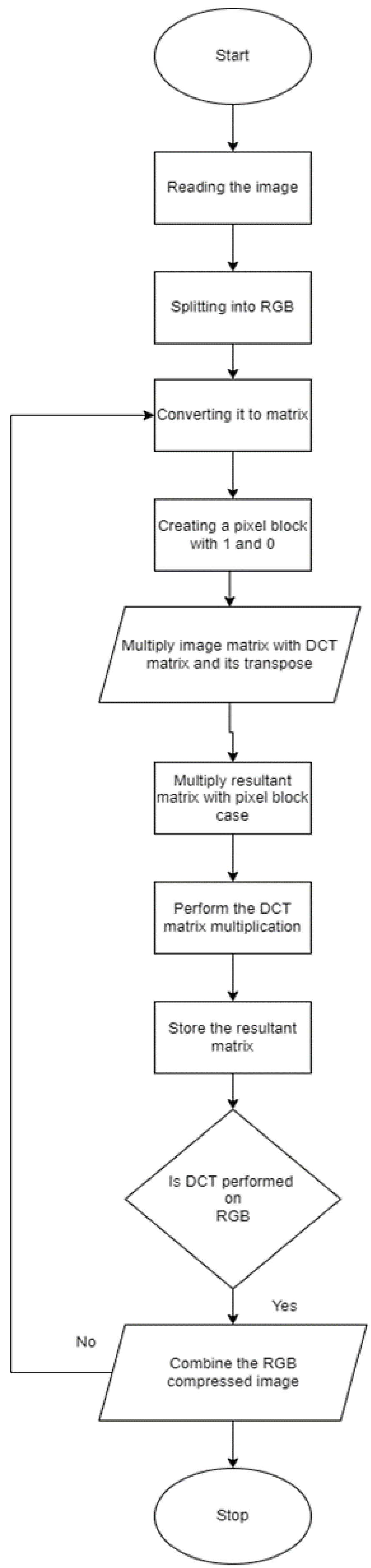
Integration of Masking Techniques:

Masking matrices were integrated into our compression algorithm to selectively retain essential image information. Through experimentation and optimization, we determined the optimal masking values to achieve maximum compression efficiency without sacrificing image quality.



Compression Using DCT Algorithm:

In our implementation, we employed the discrete cosine transform (DCT) algorithm for image compression. By dividing the image into 8 x 8 pixel blocks and applying the DCT matrix provided by MATLAB's image processing toolbox, we computed DCT coefficients. These coefficients were then multiplied with the masking matrix to retain only the essential information, resulting in a compressed image matrix with higher energy coefficients concentrated in the upper left corner of the DCT matrix.


Start
Reading the image
Splitting into RGB
Converting it to matrix
Creating a pixel block with 1 and 0
Multiply image matrix with DCT matrix and its transpose
Multiply resultant matrix with pixel block case
Perform the DCT matrix multiplication
Store the resultant matrix
Is DCT performed on RGB
Yes
No
Combine the RGB compressed image
Stop

Evaluation and Validation:

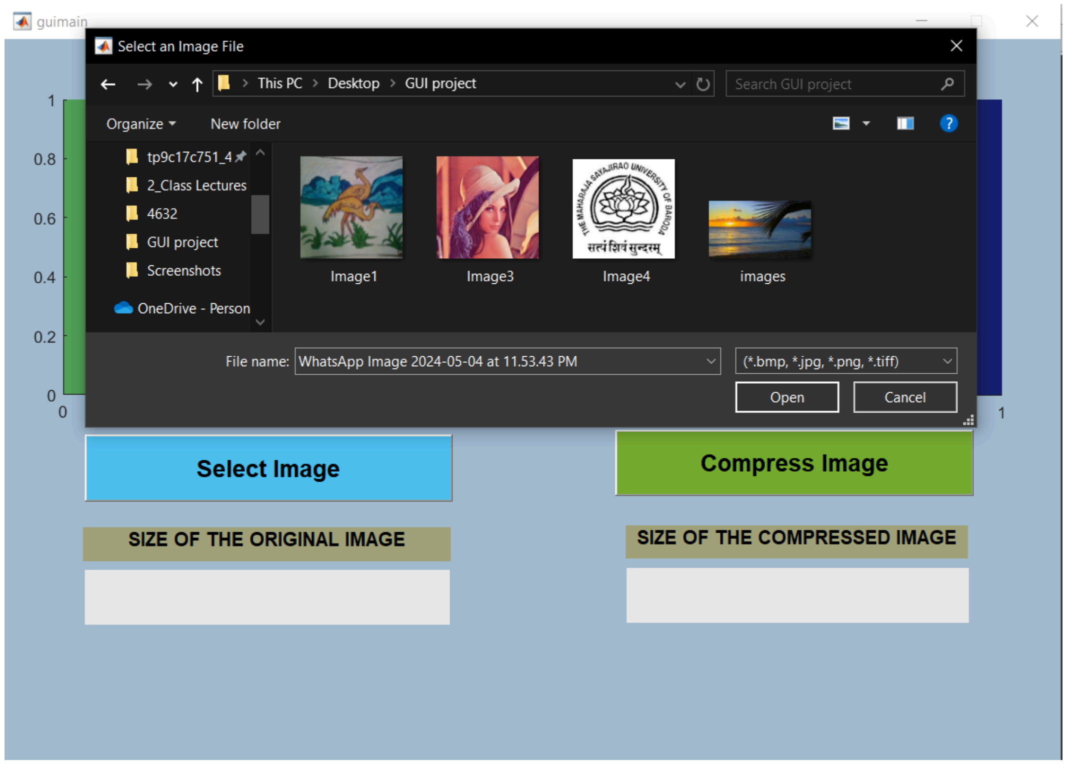
We rigorously evaluated the performance of our compression algorithm using standard image quality metrics such as peak-signal-noise ratio (PSNR) and mean-square error (MSE). Additionally, subjective quality assessment by human observers was conducted to validate the perceptual quality of the compressed images. The compression efficiency was analyzed across different image classes to assess the adaptability and effectiveness of our approach.

Integration with MATLAB GUI:

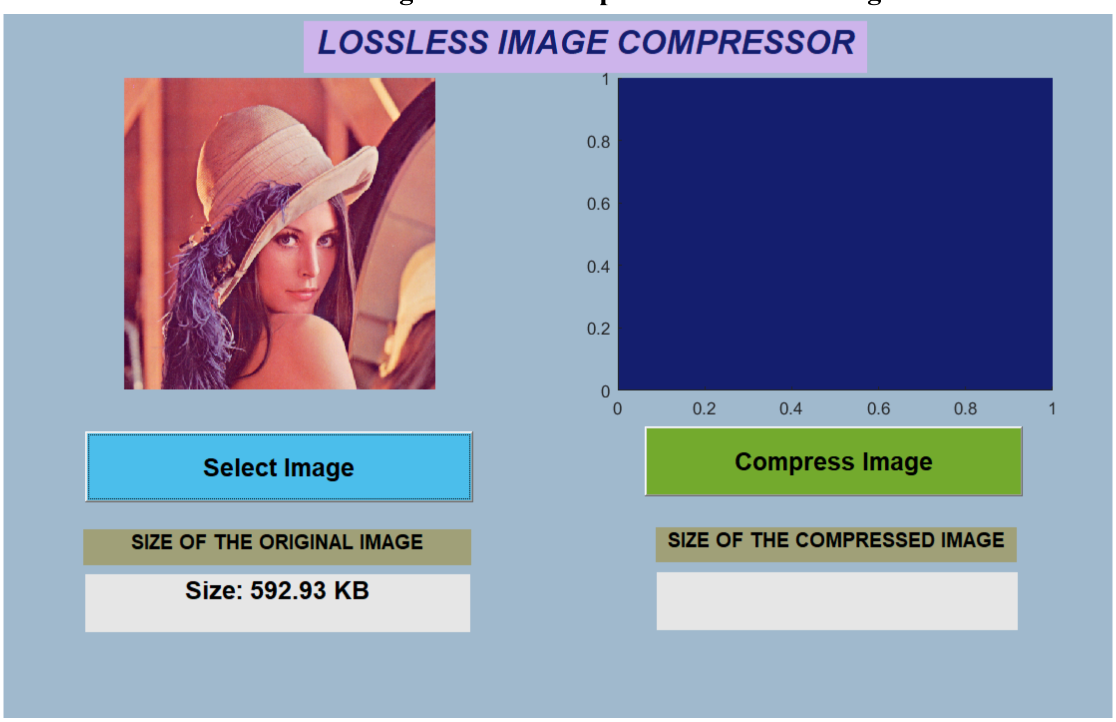
The final phase of our project involved the seamless integration of our lossless image compression algorithm with the MATLAB GUI. Users were able to access the compression tool directly through the GUI interface, simplifying the compression process and enhancing user experience. The GUI served as a platform for showcasing the capabilities of our compression algorithm to a wider audience.

Result:

1. **Select Image:**



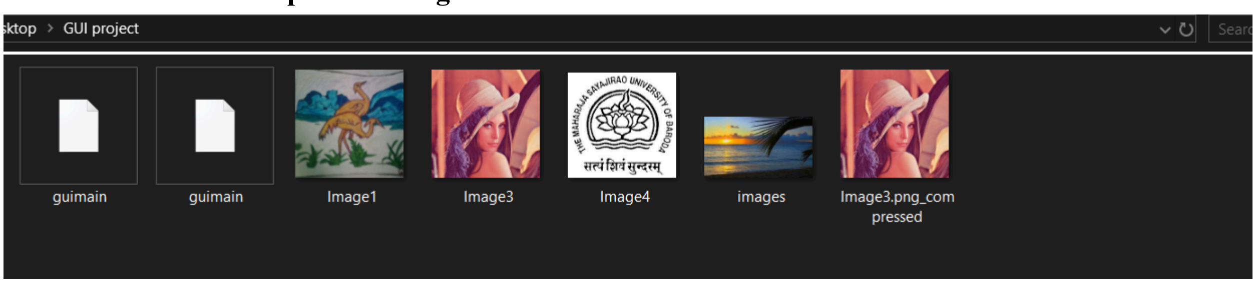
2. **Picture selected and the original size of the picture is also showing:**



3. **Pressing the compress Image results in:**

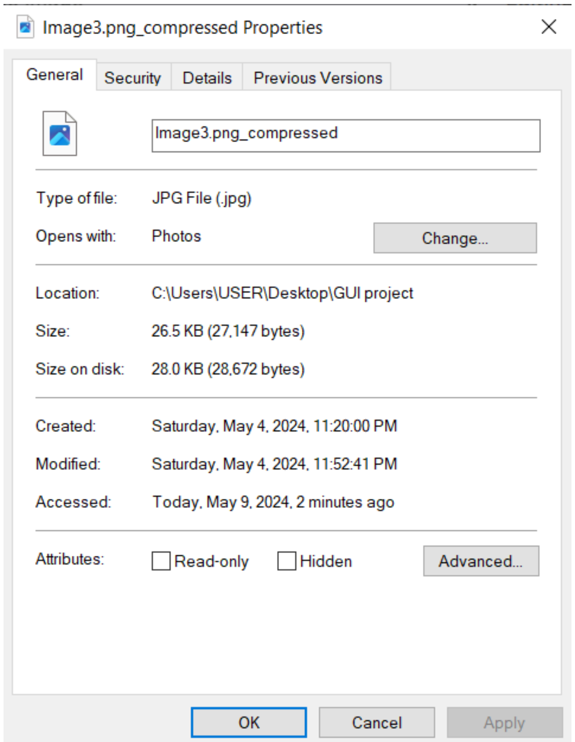
We can see, the size of the compressed image is shown here. The quality of the image is also unchanged.

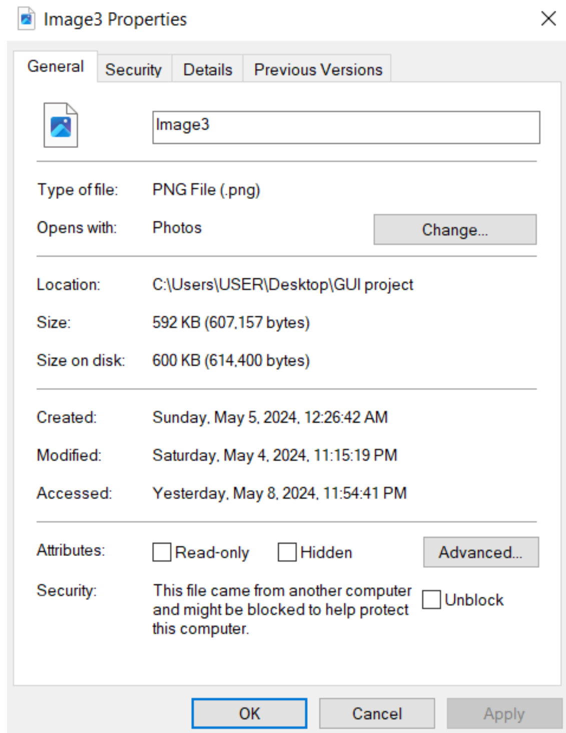
4. **Autosaved compressed image:**



Here , the last picture shows the compressed one.

5. **Compressed properties:**





Conclusion:

In summary, our project to develop a lossless image compression tool using MATLAB has been a success. We've created an algorithm that reduces file sizes while keeping image quality intact. By combining masking matrices and discrete cosine transforms, we've made compression easy and effective. Our work is a step forward in making image compression accessible to everyone, and we're excited to see how it'll benefit various fields.

Contribution of the team members:

- Sadman Shafi Ifty- Conceptual analysis and scratching the code.
- K M Istiaque- Conceptual analysis and scratching the code.
- Shafayet Sadik Sowad- Code implementation and debugging.
- MD Wahiduzzaman- User interface handling and App design.